

Geometric Visualization of the Pentagonal Number Theorem

The exceptional partitions in Euler's involution are the two staircase families
 $(2m - 1, 2m - 2, \dots, m)$ and $(2m, 2m - 1, \dots, m + 1)$.

$$(2m - 1, 2m - 2, \dots, m)$$

$$\text{size} = \frac{m(3m - 1)}{2}$$

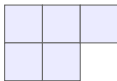
$m = 1$



(1)

$$\frac{1(3 \cdot 1 - 1)}{2} = 1$$

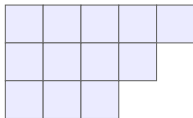
$m = 2$



(3, 2)

$$\frac{2(3 \cdot 2 - 1)}{2} = 5$$

$m = 3$



(5, 4, 3)

$$\frac{3(3 \cdot 3 - 1)}{2} = 12$$

$$(2m, 2m - 1, \dots, m + 1)$$

$$\text{size} = \frac{m(3m + 1)}{2}$$

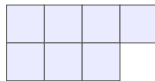
$m = 1$



(2)

$$\frac{1(3 \cdot 1 + 1)}{2} = 2$$

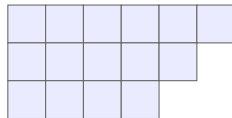
$m = 2$



(4, 3)

$$\frac{2(3 \cdot 2 + 1)}{2} = 7$$

$m = 3$



(6, 5, 4)

$$\frac{3(3 \cdot 3 + 1)}{2} = 15$$